

Customer Shopping Behavior Analysis

A Comprehensive End-to-End Data Analytics Project



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1 Executive Summary

This project analyzes the shopping behavior of 3,900 retail customers to uncover actionable insights regarding demographics, product performance, and promotional effectiveness. By building an end-to-end data pipeline—utilizing Python for data cleaning and feature engineering, SQL for exploratory data analysis, and Power BI for interactive visualization—this report identifies key drivers of customer loyalty and provides strategic recommendations to optimize marketing spend and inventory management.

2 Business Problem Statement

A leading retail company aims to better understand its customers' shopping behavior to improve sales, customer satisfaction, and long-term loyalty. With shifting purchasing patterns across demographics, product categories, and sales channels, management requires a data-driven approach to uncover which factors (such as discounts, seasons, or payment preferences) drive consumer decisions and repeat purchases.

Objective: Leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies.

3 Data Preparation & Modeling (Python)

The raw dataset (`customer_shopping_behavior.csv`) contained 3,900 records and 18 attributes. To ensure high data integrity for the SQL database, the data was cleaned and transformed using Python (`pandas`).

3.1 Loading and Initial Inspection

The data was imported and initial summary statistics and structural information was generated to understand the data types and distribution.

```
In [1]: # Loading the dataset using pandas

import pandas as pd

df = pd.read_csv('customer_shopping_behavior.csv')

In [2]: df.head()
```

Figure 1: Loading Data

Out[2]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchase
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	

Figure 2: Output of dataframe

3.2 Handling Missing Values

The initial inspection revealed that the **Review Rating** column contained 37 missing values. Rather than dropping records or filling missing values with a global average, a localized imputation strategy was used based on product category.

```
In [6]: # Imputing missing values in Review Rating column with the median rating of the product category
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

Figure 3: Data Imputation Strategy Code

```
In [7]: df.isnull().sum()
Out[7]: Customer ID      0
Age      0
Gender    0
Item Purchased  0
Category  0
Purchase Amount (USD)  0
Location  0
Size      0
Color     0
Season    0
Review Rating  0
Subscription Status  0
Shipping Type  0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method  0
Frequency of Purchases  0
dtype: int64
```

Figure 4: Output after Imputation

3.3 Data Standardization

To ensure seamless integration with the PostgreSQL database, all column headers were formatted to standard SQL naming conventions (snake_case).

```
In [8]: # Renaming columns according to snake casing for better readability and documentation
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

Figure 5: Column Standardization Code

```
In [9]: df.columns
Out[9]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
              'purchase_amount', 'location', 'size', 'color', 'season',
              'review_rating', 'subscription_status', 'shipping_type',
              'discount_applied', 'promo_code_used', 'previous_purchases',
              'payment_method', 'frequency_of_purchases'],
              dtype='object')
```

Figure 6: Output of formatted column names

3.4 Feature Engineering & Redundancy Checks

New variables were derived from existing data to enable deeper segmentation in Power BI and SQL. Specifically, continuous age data was grouped into quartiles, and text-based frequencies were mapped to numerical days. Finally, a redundancy check was performed to streamline the dataset.

```

In [10]: # create a new column age_group
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)

# create new column purchase_frequency_days

frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)

```

Figure 7: Feature Engineering and Deduplication Code

3.5 Database Integration

The fully cleaned and engineered DataFrame was pushed directly to a local PostgreSQL server using SQLAlchemy, establishing the foundation for exploratory data analysis.

Connecting Python script to PostgreSQL

```

In [26]: !pip install psycopg2-binary sqlalchemy

Requirement already satisfied: psycopg2-binary in c:\users\kiit\anaconda3\lib\site-packages (2.9.10)
Requirement already satisfied: sqlalchemy in c:\users\kiit\anaconda3\lib\site-packages (1.4.22)
Requirement already satisfied: greenlet==0.4.17 in c:\users\kiit\anaconda3\lib\site-packages (from sqlalchemy) (1.1.1)
Note: you may need to restart the kernel to use updated packages.

In [28]: from sqlalchemy import create_engine

# Step 1: Connect to PostgreSQL
# Replace placeholders with your actual details
username = "postgres" # default user
password = "mihi" # the password you set during installation
host = "localhost" # if running locally
port = "5432" # default PostgreSQL port
database = "customer_behavior" # the database you created in pgAdmin

engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

# Step 2: Load DataFrame into PostgreSQL
table_name = "customer" # choose any table name
df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"Data successfully loaded into table '{table_name}' in database '{database}'.")

Data successfully loaded into table 'customer' in database 'customer_behavior'.

```

Figure 8: Loading Data to PostgreSQL Code

4 Exploratory Data Analysis & Insights (SQL)

With the data structured in PostgreSQL, exploratory queries were executed to answer strategic business questions across core categories.

4.1 Customer Demographics & Loyalty Segmentation

Query 1: Revenue by Gender

This query identifies which demographic holds the most purchasing power, allowing the marketing team to allocate ad spend more efficiently.

```
1 SELECT
2     gender ,
3     SUM(purchase_amount) AS revenue
4 FROM customer
5 GROUP BY gender;
```

Listing 1: Total revenue generated by male vs. female customers

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Figure 9: Result: Revenue by Gender

Business Conclusion

Male customers generate significantly more revenue (\$157,890) than female customers (\$75,191), indicating a strong male-dominated purchasing demographic that should heavily influence product assortment and ad targeting.

Query 2: Revenue by Age Group

This query highlights the company's core audience, dictating the tone, style, and placement of future marketing campaigns.

```
1 SELECT
2     age_group ,
3     SUM(purchase_amount) AS total_revenue
4 FROM customer
5 GROUP BY age_group
6 ORDER BY total_revenue DESC;
```

Listing 2: Revenue contribution of each age group

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Figure 10: Result: Revenue by Age Group

Business Conclusion

Young Adults (\$62,143) and Middle-aged customers (\$59,197) are the primary revenue drivers, highlighting a younger core audience for targeted marketing campaigns.

Query 3: Customer Loyalty Segmentation

This Common Table Expression (CTE) measures the health of customer retention by grouping customers into New, Returning, and Loyal tiers.

```

1 WITH customer_type AS (
2     SELECT
3         customer_id,
4         previous_purchases,
5         CASE
6             WHEN previous_purchases = 1 THEN 'New'
7             WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
8             ELSE 'Loyal'
9         END AS customer_segment
10    FROM customer
11 )
12 SELECT
13     customer_segment,
14     COUNT(*) AS "Number of Customers"
15 FROM customer_type
16 GROUP BY customer_segment;

```

Listing 3: Customer segmentation based on previous purchases

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Figure 11: Result: Customer Segmentation Counts

Business Conclusion

The vast majority of the customer base (3,116 customers) falls into the 'Loyal' tier. While this shows excellent retention, the very low number of 'New' customers (83) signals a critical weakness in top-of-funnel customer acquisition.

4.2 Loyalty & Subscription Value

Query 4: Subscriber vs. Non-Subscriber Value

This determines the financial ROI of the subscription program by proving whether subscribers actually spend more per transaction.

```

1 SELECT
2     subscription_status,
3     COUNT(customer_id) AS total_customers,
4     ROUND(AVG(purchase_amount), 2) AS avg_spend,
5     ROUND(SUM(purchase_amount), 2) AS total_revenue
6 FROM customer
7 GROUP BY subscription_status
8 ORDER BY total_revenue DESC, avg_spend DESC;

```

Listing 4: Average spend and total revenue by subscription status

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	No	2847	59.87	170436.00
2	Yes	1053	59.49	62645.00

Figure 12: Result: Subscription ROI and Value

Business Conclusion

Non-subscribers generate a slightly higher average order value (\$59.87) than active subscribers (\$59.49) and account for the bulk of total revenue. This suggests the current subscription model is failing to drive larger cart sizes and needs optimization.

Query 5: Repeat Buyer Subscription Rate

This query identifies missed opportunities by finding highly active buyers who have not yet opted into a subscription.

```

1 SELECT
2     subscription_status,
3     COUNT(customer_id) AS repeat_buyers
4 FROM customer
5 WHERE previous_purchases > 5
6 GROUP BY subscription_status;

```

Listing 5: Subscription status of repeat buyers (5+ purchases)

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Figure 13: Result: Repeat Buyers Subscription Rate

Business Conclusion

A massive portion of highly active repeat buyers (2,518 customers with 5+ purchases) are not subscribed. This represents a major missed opportunity for loyalty program conversion campaigns.

4.3 Product Performance & Preferences

Query 6: Top Products per Category

Using a window function, this query gives the merchandising team a clear picture of their "hero" products that should always remain in stock.

```

1 WITH item_counts AS (
2     SELECT
3         category,
4         item_purchased,
5         COUNT(customer_id) AS total_orders,
6         ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id)
7         DESC) AS item_rank
8     FROM customer
9     GROUP BY category, item_purchased
10 )
11 SELECT
12     item_rank,
13     category,
14     item_purchased,
15     total_orders
16 FROM item_counts
17 WHERE item_rank <= 3;

```

Listing 6: Top 3 most purchased products within each category

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Figure 14: Result: Top 3 Products per Category

Business Conclusion

Blouses, Pants, and Jewelry are the highest-volume items across their respective categories. These "hero" products must be prioritized in supply chain logistics to prevent costly stock-outs.

Query 7: Highest Rated Products

This highlights products with high customer satisfaction, which should be heavily promoted to build brand trust.

```

1 SELECT
2     item_purchased,
3     ROUND(AVG(review_rating::numeric), 2) AS "Average Product Rating"
4 FROM customer
5 GROUP BY item_purchased
6 ORDER BY AVG(review_rating) DESC
7 LIMIT 5;
```

Listing 7: Top 5 products with the highest average review rating

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Figure 15: Result: Top 5 Highest Rated Products

Business Conclusion

Gloves (3.86) and Sandals (3.84) lead the catalog in customer satisfaction. Highlighting these highly-rated items in digital ads can be an effective way to build initial brand trust with new shoppers.

Query 8: Inventory Optimization

Crucial for supply chain management, this query identifies the exact product variants that sell the highest volume.

```

1 SELECT
2     category,
3     size,
4     color,
5     COUNT(customer_id) AS items_sold
6 FROM customer
7 GROUP BY category, size, color
8 ORDER BY items_sold DESC
9 LIMIT 10;

```

Listing 8: Most popular size and color combinations

	category text	size text	color text	items_sold bigint
1	Clothing	M	Orange	44
2	Clothing	M	Violet	42
3	Clothing	M	Teal	41
4	Clothing	M	Silver	39
5	Clothing	M	Peach	37
6	Clothing	M	White	37
7	Accessori...	M	Olive	35
8	Clothing	M	Red	34
9	Clothing	M	Charco...	34
10	Clothing	M	Gray	33

Figure 16: Result: Top Size and Color Combinations

Business Conclusion

Medium (M) sized clothing in vibrant colors like Orange, Violet, and Teal absolutely dominate sales volume, providing clear, data-driven directives for inventory procurement.

4.4 Operations, Shipping & Payments

Query 9: Shipping Type vs. Average Spend

Reveals if customers who buy premium carts are also willing to pay for faster shipping.

```

1 SELECT
2     shipping_type ,
3     ROUND(AVG(purchase_amount), 2) AS avg_purchase
4 FROM customer
5 WHERE shipping_type IN ('Standard','Express')
6 GROUP BY shipping_type;
```

Listing 9: Average Purchase Amounts between Standard and Express Shipping

	shipping_type text	avg_purchase numeric
1	Standard	58.46
2	Express	60.48

Figure 17: Result: Shipping Type vs Spend

Business Conclusion

Customers opting for Express shipping have a higher average order value (\$60.48) compared to Standard shipping (\$58.46), indicating that buyers spending premium amounts prefer and expect faster delivery options.

Query 10: Payment Method Processing Volume

Helps the finance department understand transaction volumes to negotiate better gateway fees.

```

1 SELECT
2     payment_method,
3     COUNT(customer_id) AS usage_count,
4     SUM(purchase_amount) AS total_processed
5 FROM customer
6 GROUP BY payment_method
7 ORDER BY usage_count DESC;

```

Listing 10: Usage count and total revenue by payment method

	payment_method text	usage_count bigint	total_processed numeric
1	PayPal	677	40109
2	Credit Card	671	40310
3	Cash	670	40002
4	Debit Card	636	38742
5	Venmo	634	37374
6	Bank Transfer	612	36544

Figure 18: Result: Payment Method Popularity

Business Conclusion

PayPal, Credit Cards, and Cash are practically tied as the top payment methods (each processing roughly \$40,000). E-commerce and POS checkout flows must remain heavily optimized for these primary gateways.

4.5 Discount Impact & Psychology

Query 11: High-Value Discount Shoppers

Proves that strategic discounting can successfully incentivize customers to add more items to their cart.

```

1 SELECT
2     customer_id,
3     purchase_amount
4 FROM customer
5 WHERE discount_applied = 'Yes'
6     AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);

```

Listing 11: Customers using a discount who spent above average

	customer_id bigint	purchase_amount bigint
57	116	94
58	117	86
59	120	76
60	121	84
61	122	72
62	124	89
63	125	86
64	128	89
65	130	67
66	134	65
67	135	94
68	136	81
69	137	86
70	140	95
71	146	95
72	148	64
73	149	90

Figure 19: Result: Above-Average Discount Shoppers

Business Conclusion

Discounts successfully drive high-value carts. A significant number of shoppers using discounts are spending in the \$80–\$95 range, which is well above the overall average order value of ~\$59. This proves promotions successfully incentivize customers to add more to their carts.

Query 12: Discount-Reliant Products

Identifies items that struggle to sell at full retail price, signaling a need for a permanent price reduction.

```

1 SELECT
2     item_purchased,
3     ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) /
4         COUNT(*), 2) AS discount_rate
5 FROM customer
6 GROUP BY item_purchased
7 ORDER BY discount_rate DESC
8 LIMIT 5;

```

Listing 12: Products with the highest percentage of discount purchases

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Figure 20: Result: Most Discount-Reliant Products

Business Conclusion

Hats (50.00%) and Sneakers (49.66%) are the most discount-reliant products in the catalog. Because nearly half of their sales require a promotion to convert, these items may be overpriced at their base retail rate and require permanent price adjustments.

Query 13: Discount Impact on Customer Satisfaction

Analyzes buyer psychology to see if discounted items are perceived as great deals or cheap quality.

```

1 SELECT

```

```

2  discount_applied,
3  ROUND(AVG(review_rating::numeric), 2) AS avg_rating,
4  COUNT(customer_id) AS total_reviews
5 FROM customer
6 GROUP BY discount_applied;

```

Listing 13: Average ratings for discounted vs. non-discounted items

	discount_applied text	avg_rating numeric	total_reviews bigint
1	No	3.76	2223
2	Yes	3.74	1677

Figure 21: Result: Discount Impact on Ratings

Business Conclusion

Discounts have virtually no psychological impact on customer satisfaction or perceived quality. Customers rate full-price items (3.76) almost identically to discounted items (3.74).

4.6 Geography & Seasonality

Query 14: Top Revenue-Generating Locations

Tells the marketing team exactly where to target regional digital ads and physical storefronts.

```

1 SELECT
2   location,
3   COUNT(customer_id) AS total_orders,
4   SUM(purchase_amount) AS total_revenue
5 FROM customer
6 GROUP BY location
7 ORDER BY total_revenue DESC
8 LIMIT 5;

```

Listing 14: Top 5 locations by total revenue

	location text	total_orders bigint	total_revenue numeric
1	Montana	96	5784
2	Illinois	92	5617
3	California	95	5605
4	Idaho	93	5587
5	Nevada	87	5514

Figure 22: Result: Top Revenue Locations

Business Conclusion

Montana is the surprising top-performing market, leading in both order volume and total revenue (\$5,784). It slightly outperforms traditionally massive markets like Illinois and California, revealing a strong regional hotspot for targeted ad spend.

Query 15: Seasonal Purchasing Behavior

Maps out the annual retail cycle, allowing the company to plan major sales events around seasonal peaks and valleys.

```

1 SELECT
2     season,
3     COUNT(customer_id) AS total_orders,
4     SUM(purchase_amount) AS total_revenue,
5     ROUND(AVG(purchase_amount), 2) AS avg_order_value
6 FROM customer
7 GROUP BY season
8 ORDER BY total_revenue DESC;
```

Listing 15: Purchasing behavior across different seasons

	season text	total_orders bigint	total_revenue numeric	avg_order_value numeric
1	Fall	975	60018	61.56
2	Spring	999	58679	58.74
3	Winter	971	58607	60.36
4	Summer	955	55777	58.41

Figure 23: Result: Seasonal Order Value and Volume

Business Conclusion

Fall is the definitive profit driver, generating the highest total revenue (\$60,018) and the highest Average Order Value (\$61.56). Spring captures the highest raw foot traffic (999 orders), while Summer represents a significant seasonal slump across all performance metrics.

5 Interactive Visualization (Power BI)

A Power BI dashboard was developed to allow stakeholders to interactively explore the SQL findings. The dashboard utilizes global slicers to enable deep-dive filtering across key business metrics.

5.1 Executive Summary Dashboard

This view provides a high-level assessment of business health, featuring primary KPI cards (Total Revenue, Active Customers, Average Order Value) and clear breakdowns of sales by category and age group. A cohesive color palette was implemented to provide immediate visual hierarchy and professional polish.

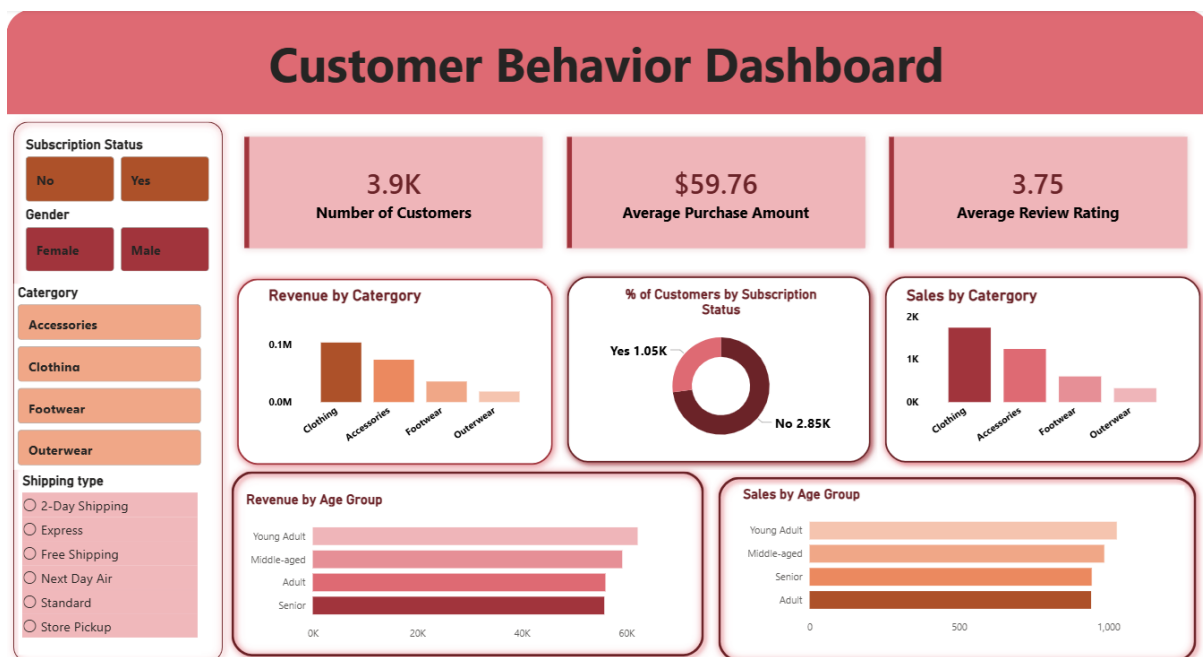


Figure 24: Power BI Executive Summary Dashboard featuring key business metrics.

6 Strategic Suggestions & Recommendations

Based on the end-to-end data analysis across Python, SQL, and Power BI, the following data-driven actions are highly recommended to the executive team:

- Capitalize on the Fall Season (Premium Purchasing):** Since customers are willing to spend more per transaction in the Fall (highest AOV of \$61.56), marketing should focus heavily on upselling premium items and promoting higher-margin products during these months.
- Maximize Spring Traffic:** Because order volume peaks in the Spring (999 orders), this is the optimal time to run "Bundle and Save" promotions to force the Average Order Value upward while capturing the high seasonal foot traffic.
- Targeted Inventory Procurement:** Procurement teams should prioritize manufacturing and stocking the specific size and color combinations (e.g., Medium, vibrant colors like Orange and Violet) that historically drive the most volume, actively avoiding overstock on low-ranking variants.

- **Subscription Revamp & Optimization:** Data indicates a massive missed opportunity: 2,518 highly active repeat buyers are not subscribed. Furthermore, current subscribers do not spend more than non-subscribers. The business must restructure the loyalty program benefits and aggressively market it to the 'Returning' customer tier.
- **Address the Summer Slump:** To correct the massive dip in Summer sales, the company should introduce aggressive mid-summer clearance events or exclusive loyalty-member flash sales to drive traffic during these slower months.
- **Re-Evaluate "Discount-Reliant" Items:** Hats and Sneakers require a discount to sell nearly 50% of the time. The merchandising team should consider a permanent price reduction on these items rather than relying on constant promotional crutches.

7 Final Conclusion

This Comprehensive Data Analytics Project successfully transformed raw, unstructured retail data into a clear, strategic business asset. By leveraging **Python** for robust data preprocessing and feature engineering, **PostgreSQL** for deep-dive exploratory data analysis, and **Power BI** for dynamic, interactive visualization, this report uncovered critical inefficiencies and high-value opportunities within the customer lifecycle.

The data clearly dictates that while the company possesses an incredibly loyal customer base, it suffers from weak top-of-funnel acquisition (low 'New' customer volume) and an under-optimized subscription model. However, by leaning into geographical hotspots like Montana, aligning inventory procurement with the dominant "Medium/Vibrant" product variants, and shifting marketing budgets to capture the high-value Fall shopper, the company is exceptionally well-positioned to maximize revenue margins and drive sustainable, long-term retail growth.